B.TECH CSE (AI & ML) · SEM V · OPERATING SYSTEMS AND SECURITY

# Class Test · Set 1

Total questions: 26 Maximum marks: 25 Duration: _____

Name: Roll No.: Date:

**Instructions**

- Section A: attempt all 20 MCQs (0.5 marks each). Section B: attempt all 4 questions (2.5 marks each). Section C: attempt any 1 of 2 questions (5 marks).
- Base-and-bound: the bound is the size of the address space, so a virtual address is valid only if $0 \le VA <$ bound.
- Turnaround = completion − arrival; waiting = turnaround − burst; response = first run − arrival. Context-switch time is zero.
- Scheduling ties are broken by earlier arrival, then lower process number. In Round Robin, a newly arriving process joins the queue before a process preempted at the same instant.
- Show all working for numericals; draw Gantt charts with times marked.

## Section A · Multiple Choice

20 × 0.5 = 10 marks · attempt all

**A1** Which OS idea gives every running process the illusion of having its own CPU, even though one CPU is shared?

**A** Abstraction
**B** Caching
**C** Virtualization
**D** Spooling

**A2** A file, which hides disk sectors and blocks behind a name and a byte stream, is an example of:

**A** Abstraction
**B** Virtualization
**C** Paging
**D** Preemption

**A3** Which statement about containers on a Linux host is correct?

**A** Each container boots its own kernel
**B** All containers share the host's kernel
**C** Containers need a type-1 hypervisor to run
**D** Containers cannot isolate file systems

**A4** Which instruction can execute only in kernel mode?

**A** Adding two registers
**B** Loading a value from the process's own stack
**C** Calling a library function such as strlen()
**D** Disabling interrupts

**A5** A program calls `read()` on a file. Which order of events is correct?

**A** Library wrapper → trap → kernel handler → return-from-trap
**B** Kernel handler → trap → library wrapper → return
**C** Trap → library wrapper → kernel handler → return
**D** Library wrapper → kernel handler, with no change of CPU mode

**A6** Which of the following is NOT stored in a process's PCB?

**A** Process state
**B** Saved register values
**C** List of open files
**D** The data stored on the process's heap

**A7** In the 5-state model, a process that has been created but not yet admitted by the OS is in which state?

**A** Ready
**B** New
**C** Blocked
**D** Running

**A8** A running process issues a disk read. Which state does it move to?

**A** Ready
**B** Terminated
**C** Blocked
**D** New

**A9** The disk read for a blocked process completes. Which state does it move to?

**A** Running
**B** Ready
**C** New
**D** Terminated

**A10** How many times is X printed?

```
int main() {
    fork();
    fork();
    fork();
    printf("X");
}
```

**A** 3
**B** 6
**C** 4
**D** 8

**A11** Which call, when it succeeds, never returns to the code that called it?

**A** fork()
**B** wait()
**C** exec()
**D** getpid()

**A12** A child has terminated but its parent has not yet called `wait()`. The child is called:

**A** A zombie process
**B** An orphan process
**C** A daemon process
**D** A thread

**A13** A process runs an infinite loop and makes no system calls. What lets the OS take the CPU back?

**A** A page fault
**B** A timer interrupt
**C** A system call
**D** A call to fork()

**A14** Switching between two processes requires changing which of these, while switching between two threads of the same process does not?

**A** Program counter
**B** Stack pointer
**C** Address-space registers (page-table base or base/bound)
**D** General-purpose registers

**A15** Three processes arrive at time 0 with bursts 10, 1 and 1 ms, in that order. What is the average waiting time under FCFS?

**A** 7 ms
**B** 1 ms
**C** 4 ms
**D** 0 ms

**A16** If the Round Robin time quantum is larger than every process's burst, Round Robin behaves like:

**A** SJF
**B** SRTF
**C** Priority scheduling
**D** FCFS

**A17** Response time of a process is defined as:

**A** Completion time − arrival time
**B** Burst time
**C** Time spent in the Blocked state
**D** Time of first run − arrival time

**A18** Which IPC mechanism involves the kernel only during setup, after which processes exchange data without the kernel copying it?

**A** Pipe
**B** Socket
**C** Shared memory
**D** Message queue

**A19** A process has base = 6000 and bound = 900. It issues virtual address 899. What happens?

**A** Physical address 6899 is accessed
**B** The MMU traps
**C** Physical address 5101 is accessed
**D** Physical address 6900 is accessed

**A20** Which hardware unit performs the base-and-bound check on every memory access?

**A** The CPU scheduler
**B** The MMU
**C** The PCB
**D** The cache controller

## Section B · Short Answer

4 × 2.5 = 10 marks · attempt all

**B1** Compare a virtual machine and a container on four points: kernel, isolation strength, start-up time and resource overhead. Give one situation where you would choose each. [2.5]

Answer

**B2** Predict the output of this program and the order in which lines appear. Explain why the child's change to x does not affect the parent. [2.5]

```
int x = 10;
int pid = fork();
if (pid == 0) {
    x = x * 2;
    printf("child %d\n", x);
} else {
    wait(NULL);
    x = x + 1;
    printf("parent %d\n", x);
}
```

Answer

**B3** A process has base = 12000 and bound = 3000 (the bound is the size, so valid VAs are 0 to 2999). For each virtual address, state valid or invalid and give the physical address where valid: (a) 0 (b) 2999 (c) 3000 (d) 1500 [2.5]

Answer

**B4** Three processes arrive at time 0 in the order P1, P2, P3 with bursts 5, 3 and 1 ms. Using Round Robin with quantum 2 ms, draw the Gantt chart and compute the waiting time and response time of each process. [2.5]

Answer

## Section C · Long Answer

attempt any 1 of 2 · 5 marks

**C1** Four processes have these arrival and burst times (ms): P1 (0, 7), P2 (2, 4), P3 (4, 1), P4 (5, 4). [5]

(a) Draw the Gantt chart for non-preemptive SJF.

(b) Draw the Gantt chart for SRTF.

(c) Compute the average waiting time for both and explain the difference.

Answer

**C2** A user types `sort data.txt` at a shell prompt. [5]

(a) Draw the 5-state process model and label every transition with the event that causes it.

(b) Trace the states the `sort` process passes through from creation to termination, given that it reads the file from disk.

(c) Explain which system calls the shell uses, why the shell calls `wait()`, and what would happen to the finished child if it did not.

Answer

— End of paper —